



秀傳醫療財團法人彰濱秀傳紀念醫院
Chang Bing Show Chwan Memorial Hospital

檢測報告

Test Report



Chang Bing Show Chwan



「RELANO WOUND SPRAY」
Skin Sensitisation Test
(Guinea Pig Maximisation Test (GPMT))

Final Report

Study Number: TW025-23021L01

Sponsor Name: Nanoplus Life Biomedical Technology Co., Ltd

Address: 9 F., No. 179, Sec. 2, Tiding Blvd., Neihu Dist., Taipei City 114735, Taiwan (R.O.C.)

Test Facility: Chang Bing Show Chwan Memorial Hospital Animal Center (CBSCMHAC)

Address: No. 6, Lugong Rd., Lukang Township, Changhua County 505, Taiwan (R.O.C.)

Test Facility Approval

Quality Assurance Auditor

Chia Yun Tsai

Signer Name: Chia-Yun Tsai, MS

Aug. 22, 2023

Date

Study Director

De Wei Lai

Signer Name: De-Wei Lai, Ph.D.

Aug. 22, 2023

Date

Facility Manager

Chi-Jung Yeh

Signer Name: Chi-Jung Yeh, MT

Aug. 22, 2023

Date

Good Laboratory Compliance Statement

This study was conducted in compliance with the Organisation for Economic Co-operation and Development (OECD) -Principles of Good Laboratory Practice (GLP) and Taiwan Food and Drug Administration - Good Laboratory Practice for Nonclinical Laboratory Studies.

The laboratory is exempt from the following provisions: Sections 6.2.2, 6.2.3, 6.2.4, 6.2.5 in Taiwan Food and Drug Administration -Good Laboratory Practice for Nonclinical Laboratory Studies and The Organization for Economic Co-operation and Development (OECD) -Principles of Good Laboratory Practice (GLP). Characterization and verification of the test item identity, strength, purity, stability, composition, method of synthesis, fabrication, derivation or other characteristics provided from sponsor, which are the responsibility of the sponsor.

Study Director

De Wei Lai

Signer Name: De-Wei Lai, Ph.D.

Aug. 22.2023

Date

Quality Assurance Statement

Study Number TW025-23021L01

Study Name 「RELANO WOUND SPRAY」
Skin Sensitisation Test (Guinea Pig Maximisation Test (GPMT))

This study has been divided into several phases for inspection. Using a random sampling approach, each of these phases was monitored by quality assurance representative over a series of inspections. Related procedures, equipment records, and documentation are examined in order to assure that the study is performed in accordance with GLP guidelines from the regulatory body, such as the Taiwan Food and Drug Administration and OECD.

The following list describes the inspection date, inspection type, inspection phase, and report dates of inspections conducted by Quality Assurance in this study.

INSPECTION DATE	INSPECTION TYPE	STUDY PHASE	DATE REPORTED TO STUDY DIRECTOR	DATE REPORTED TO FACILITY MANAGER
Jun. 21, 2023	Study-Based inspections	Draft Study Plan Review & Operator's Qualification	Jun. 21, 2023	Jun. 21, 2023
Jun. 27, 2023	Study-Based inspections	Preparation of Extracts (Induction Phase I)	Jun. 27, 2023	Jun. 27, 2023
Aug. 17, 2023	Study-Based inspections	Raw Data Review	Aug. 17, 2023	Aug. 17, 2023
Aug. 17, 2023	Study-Based inspections	Final Report Review	Aug. 17, 2023	Aug. 17, 2023

I accept the responsibility for conducting inspection to assure that the methods and procedures used in the study and the reported results accurately reflect the raw data of the study.

Quality Assurance Auditor

Chia Yun Tsai

Signer Name: Chia-Yun Tsai, MS

Aug. 22, 2023

Date

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Summary

The test item 「RELANO WOUND SPRAY」 was evaluated for the potential to cause delayed dermal contact sensitization in a guinea pig maximization test. This study was conducted based on the requirements of ISO 10993-10 (2021), Biological evaluation of medical devices - Part 10: Tests for skin sensitization (2021). The test item was extracted in 0.9% saline and sesame oil. Each extract was intradermally injected and occlusively patched to ten test guinea pigs (per extract). Following a recovery period, the test and control animals received a challenge patch of the appropriate test item extract, the vehicle control. All sites were scored for dermal reactions at 24 and 48 hours after patch removal.

The test item extracts showed no evidence of causing delayed dermal contact sensitization in the guinea pig. The test item was not considered a sensitizer in the guinea pig maximization test.

1. Objectives

The purpose of this study was evaluating the potential of the test items to cause delayed dermal contact sensitization in the guinea pig maximization test.

2. Test Facility

2.1.Name:

Chang Bing Show Chwan Memorial Hospital Animal Center (CBSCMHAC)

2.2.Address:

1F., No. 6, Lugong Rd., Lukang Township, Changhua Country 505, Taiwan (R.O.C.)

3. Personnel

Associates involved in this study was appropriately qualified and trained.

3.1.Study Director:

De-Wei Lai, Ph.D.

Phone: +886-4-7813888#73630

E-mail: deweilai123@gmail.com

Address: 3F., No. 6, Lugong Rd., Lukang Township, Changhua County 505, Taiwan (R.O.C.)

3.2.Study Personnel:

De-Wei Lai, Ph.D.

Hao-Ting Wu, MS

Wei-Jen Hsu, BS

3.3.Study Location:

Operation room

Preparation of extracts and skin sensitisation test

Rat room

Animal feeding

4. Sponsor Information

4.1.Name:

Nanoplus Life Biomedical Technology Co., Ltd

4.2.Address:

9 F., No. 179, Sec. 2, Tiding Blvd., Neihu Dist., Taipei City 114735, Taiwan (R.O.C.)

4.3.Sponsor's Representative and contact information:

Tzu-Ying Hung (洪慈營)

Phone: (02)2656-0555

Email: Cecilia@nanoplustech.com

5. Study Schedule

1. Study initiation date:	Jun. 26, 2023
2. Test item receipt date:	Jun. 16, 2023
3. Experimental starting date:	Jun. 27, 2023
4. Experimental completion date:	Jul. 24, 2023
5. Study completion date:	The date Study Director signed in the Final Report

6. Test Item

6.1. Test Item:

Name of Test Item: RELANO WOUND SPRAY

Receipt Date: Jun. 16, 2023

Lot Number: LOT51223050003

Test Item No.: TW025-23021L01-LOT51223050003 (1, 2, 3): Polar Extracts

TW025-23021L01-LOT51223050003 (4, 5, 6): Non-Polar Extracts

TW025-23021L01- LOT51223050003 (7, 8, 9): Polar Extracts (retention)

TW025-23021L01-LOT51223050003 (10, 11, 12): Non-Polar Extracts (retention)

Type: Medical Devices , Category : Class1; Surface Device

External Features: Liquid

Surface Area: Not Available

Thickness: Not Available

Concentration: N/A

Purity: N/A

Expiry Date: May 26, 2025

Manufacture Date: May 26, 2023

Amount: Test: 6 bot* 50±2mL, Retention: 6 bot* 50±2mL

Major Components: Nano Ion Water, Carboxymethylcellulose

Sterilization: No

Color: Colorless

Storage Condition: Room Temperature (SC-01 region)

Cut or Not: N/A

Extraction: By Weight: (0.2±10%)g/mL; Condition: (50±2)°C (72±2)Hours

Test Conditions: No Dilution

Others: For Skin Sensitisation Test: Guinea Pig Maximisation Test; Test Item/ Control Item/ Specimen Information Form as Appendix2

7. Test System

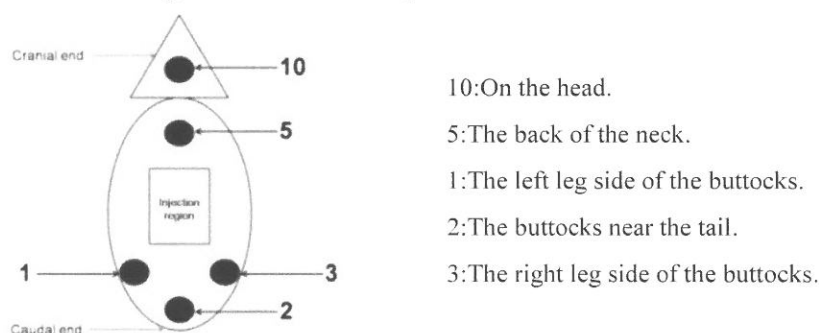
7.1. Test System:

- 7.1.1. Species: Guinea pig
- 7.1.2. Strain: Hartley
- 7.1.3. Weigh: Initial body weight 306.23 to 395.03 g.
- 7.1.4. Sex: Female healthy young adult animals were used (Non-pregnant and nulliparous).
- 7.1.5. Number of Animals: Thirty-two. Thirty for testing, two for spare.
- 7.1.6. Source: QuanXin Regenerative Biomedical Co., Ltd. (No. 145, Xingda Rd., South Dist., Taichung City 402202, Taiwan (R.O.C.)).
- 7.1.7. Feeding Region: Chang Bing Show Chwan Memorial Hospital Experimental Animal Center Rat Room. (1F., No. 6, Lugong Rd., Lukang Township, Changhua County 505, Taiwan (R.O.C.))
- 7.1.8. Acclimation Period: Quarantine in the quarantine room for 6 days and transfer to the rat room for 4 days of acclimatization.

7.1.9. Animal Facility Environment Control:

- i Temperature for the room: 17.0~23.0°C
- ii Relative humidity: 30~69.9%
- iii Light cycle: 12 hours light, 12 hours dark (set 07:00 open; 19:00 close)
- iv Condition: Animals were housed in groups in plastic suspended cages identified by a card indicating the study number, animal number, experimental group, sex, and experimental date, species, strain, and source.
- v Diet: 3023 Maintenance Diet-Guinea Pig –lot number: 202211101203 (Altromin) was supplied ad libitum throughout the study period.
- vi Bedding: Biocob Corn Cob Lab Bedding-lot number: 10530396 (1) was supplied throughout the study period.
- vii Potable water: RO and sterilization water were supplied ad libitum throughout the study period. It will be distributed to the polycarbonate bottles and autoclaved, then attached to the cages.
- viii Animal Marking and identification: Color in the corresponding positions with a marker pen as Figure 1, and the cumulative number indicated the animal number. Purple was used for polar extract testing; Black was used for Non-polar testing.

Figure 1. Annotation position of animal number



7.2. The procedures for this study were approved by the Chang Bing Show Chwan Memorial Hospital Institutional Animal Care and Use Committee (IACUC) prior to conduct. IACUC approval Number: 112024.

7.3. Selection: Only healthy, previously unused animals free of mechanical irritation or trauma that could interfere with the test were selected.

7.4. Justification of Test System: The albino guinea pig (animal) has been used historically for sensitization studies (Magnusson and Kligman, 1970). The guinea pig is believed to be the most sensitive animal model for this type of study.

8. Study Design

This test procedure refers to ISO10993-10 (2021), OECD Guideline 406 (2021) and SOP GLP-TM-10-02 Skin Sensitisation Test (Guinea Pig Maximisation Test (GPMT)).

8.1.Reagent, Materials and Equipment:

- 8.1.1. Biological Safety Cabin (BSC), model: AC2-4S9-NS, supplier: ESCO, equipment number: BSC-06, calibration date: Oct. 25, 2022
- 8.1.2. Rotary shaking incubator, model:LM-575RD, supplier: YIHDER, equipment number: SI-01, calibration date: Mar. 29, 2023
- 8.1.3. Electric shaver, model:CF-202, supplier: CHI FON SHAVER IND. CO., LTD., calibration date: N.A.
- 8.1.4. Vernier caliper, model: CD-6"CSX, equipment number: DC-01, calibration date: Aug. 23, 2022
- 8.1.5. Electronic balance, model: PGW3502E, supplier: ADAM, equipment number: BAL-01, calibration date: Mar. 29, 2023
- 8.1.6. Electronic balance, model: PA224C, supplier: OHAUS, equipment number: BAL-05, calibration date: Mar. 29, 2023
- 8.1.7. Vortex, model: VM-1000, supplier: TYPHOON, equipment number: VOX-07, calibration date: N.A.
- 8.1.8. 1000µL Pipette, supplier: Sorenson, equipment number: P1mL-06, calibration date: Mar. 29, 2023
- 8.1.9. 1000µL tip, catalog: TF112-1000, supplier: QSP, lot: 22071267, expiry date: Feb. 11, 2027
- 8.1.10. Extraction container, catalog: 5100050C, lot: 50C2G1128B, supplier: CAPP, expiry date: Aug. 31, 2024
- 8.1.11. Marker pen
- 8.1.12. 15mL tube, lot: 05222601, supplier: protech, expiry date: Feb. 26, 2025
- 8.1.13. Gauze, supplier: BELIA, lot: 20211018, expiry date: Oct. 18, 2024
- 8.1.14. Self-adhesive elastic bandage, lot: 103278623, supplier: 3M, expiry date: Nov. 05, 2025
- 8.1.15. Tubular Net Bandage, lot: 202105, supplier: I-M, expiry date: Jun. 24, 2026
- 8.1.16. Three-Way Stopcock, catalog: R1-FL-3CAPS, lot: OH13M, supplier: Top, expiry date: Jul. 31, 2025
- 8.1.17. 10mL Syringe with Needle, catalog: 302149, lot: 0233387, supplier: BD, expiry date: Aug. 31, 2025
- 8.1.18. Freund's Complete Adjuvant (FCA), catalog: F5881, lot: SLCQ7099, supplier: Sigma, expiry date: May 23, 2024
- 8.1.19. Sodium dodecyl sulfate (SLS or SDS), catalog: 1614363, lot: R11250, supplier: Sigma, expiry date: Jun. 05, 2024
- 8.1.20. 75% alcohol, supplier: TOUN-SHIN CO., LTD., lot: HA221127, expiry date: Nov. 21, 2026
- 8.1.21. Cotton swabs, supplier: BELIA, lot: 20230221-3, expiry date: Feb. 21, 2026
- 8.1.22. 1mL Syringes 26Gx1/2", supplier: PERFECT MEDICAL INDUSTRY CO., LTD. ; lot: 220304, expiry date: Feb. 12, 2025
- 8.1.23. 0.9% saline, lot: 1TKB2022, supplier: Taiwan Biotech Co., Ltd., expiry date: Aug. 01, 2025
- 8.1.24. Sesame oil, catalog: S3547, lot: MKCL0266, supplier: Sigma, expiry date: Mar. 29, 2024
- 8.1.25. Dimethyl sulfoxide (DMSO), catalog: D2650, lot: RNBK7774, supplier: Sigma, expiry date: Apr. 30, 2024

8.2.Preparation of Extracts

8.2.1. Test item was taken according to the sampling method (whole sampling added additional volume of extraction vehicle that the test sample absorbed when performing the extraction, using the weight data of the test item provided by the CBSCMHAC detection). The extraction was performed in close inert containers according to the extraction ration of $(0.2 \pm 10\%)g$: 1 mL (sample: extraction vehicle). The vehicle (without the test items) was similarly prepared to serve as the control. The extraction steps referred to SOP GLP-TM-10-03 Test Item Extraction and ISO10993-12 (2021).

8.2.1.1.The Extraction Vehicle:

Polar solvent: 0.9% saline; Non-polar solvent: Sesame oil

8.2.1.2.Extraction Condition:

$50 \pm 2^\circ\text{C}$ for 72 ± 2 hours

8.3.Test Procedure:

8.3.1. Induction phase I

8.3.1.1.The test procedure, reagent, material, equipment, and related operation time points were recorded in “GLP-TM-10-02-04T Skin Sensitisation Test Procedure-II” and referred to SOP GLP-TM-10-02 Skin Sensitisation Test (Guinea Pig Maximisation Test (GPMT)) 7.3.7.1 Program B.

8.3.2. Recorded the body weight before and after the test in “GLP-TM -10-02-02F Skin Sensitisation Test - Animal Body Weight Record”.

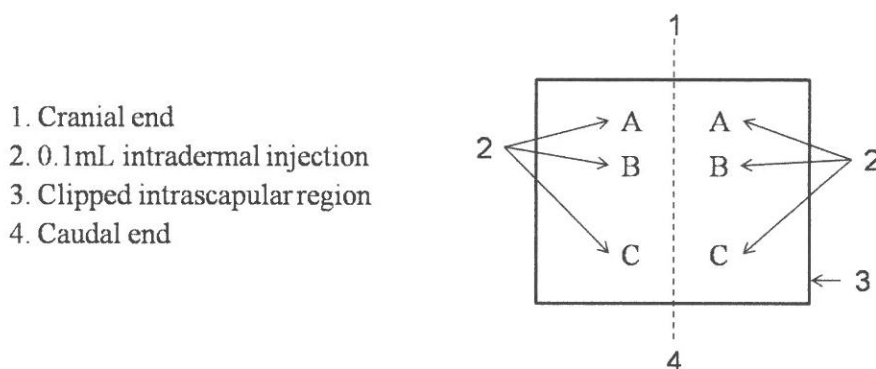
8.3.2.1.On the first day of treatment, the animals were weighed and arbitrarily assigned to a treatment group as shown below

Table 1: Treatment Group Assignment

Vehicle	Treatment Group	Number of Animals
Polar solvent: 0.9% saline	Test	10
	Control	5
Non-polar solvent: Sesame oil	Test	10
	Control	5

The fur over the dorsoscapular region was removed with an electric clipper. The test animals were injected with the test item extract and the control animals were inject with the vehicle control. Three rows of intradermal injections (two injections per row) were given to each animal within an approximate 2 cm x 4 cm boundary of the fur clip area as illustrated below

Figure 2. Location of Intradermal Injection Sites



Control Animals:

- A. 0.1 mL of 1:1 (v/v) mixture of FCA and 0.9% saline
- B. 0.1 mL of vehicle
- C. 0.1 mL of a 1:1 mixture of the 1:1 (v/v) FCA/ 0.9% saline mixture and the vehicle

Test Animals:

- A. 0.1 mL of 1:1 (v/v) mixture of FCA and 0.9% saline
- B. 0.1 mL of test extract
- C. 0.1 mL of a 1:1 mixture of the 1:1 (v/v) FCA/0.9% saline mixture and the test extract

8.3.3. Induction phase II

- 8.3.3.1. At 6 ± 1 days after completion of the Induction I injection, the fur over the dorsoscapular region (same area as used during Induction I) of each animal was removed with an electric clipper (if necessary). The area was treated with a 10% SLS to coat the skin by cotton swab. The SLS applied to provoke a mild acute inflammation, was massaged into the skin over the injection site. The area was uncovered.
- 8.3.3.2. At 24 hours (± 2 hours) after SLS treatment. An approximate 2 cm x 4 cm section of gauze patch, saturated with 0.5 mL of freshly prepared test item extract, was then topically applied to the previously injected sites of the test animals. The control animals were similarly patched with the appropriate vehicle control. Each patch was secured with a nonreactive tape and the trunk of each animal was wrapped with an elastic bandage. At 48 hours, the bandages and patches were removed.

8.3.4. Challenge phase

8.3.4.1. At 14 ± 1 days after completion of Induction phase II, the fur was removed from both sides flank areas with an electric clipper. Appropriate sites for these hairless areas were selected and applied by the patches. Gauze patches were saturated with 0.5 mL of the test item extract or control vehicle. The test item extract was applied to the right flank of each animal and the control vehicle was applied to the left flank of each animal. The trunk of each animal was wrapped with anelastic bandage to maintain well-occluded sites. At 24 hours, the wraps and gauze were removed. Any residue remaining at the sites was removed.

8.3.5. Laboratory Observations

8.3.5.1. Animals were observed daily for general health and recorded in “GLP-TM-10-04-04F Laboratory Animal Observation Record” .

8.3.5.2. Observations and take pictures for dermal reactions were conducted at 24 ± 2 and 48 ± 2 hours after challenge patch removal. Dermal reactions were scored in accordance with the criteria shown below, and recorded in “GLP-TM-10-02-05F Individual Values of Skin Reactions-II”.

Table 2: Magnusson and Kligman scale

Patch test reaction	Grading scale
No visible change	0
Discrete or patchy erythema	1
Moderate and confluent erythema	2
Intense erythema and/ or swelling	3

9. Results Analysis

9.1. Magnusson and Kligamn grades of 1.0 or greater in the individual animals of test group generally indicated sensitization, provided grades of less than 1.0 were seen on the individual animals of control animals. If grades of 1.0 or greater were noted in control animals, then the reactions of test animal which exceed the most severe reaction in control animals were presumed to be due to sensitization. The frequency of positive challenge result of the test group was more than 30%, the test result was positive. If the response was equivocal, rechallenge was recommended to confirm the results from the first challenge.

9.2. According to ISO 10993-10 (2021), the skin sensitivity test (guinea pig maximization method) conducted a positive control substance confirmed, which was performed within six months before the study.

10.Results

10.1.Extraction

10.1.1. Polar Extract

Phase	Group	Extraction Ratio	Weight of Test Item	Extraction Volume	Condition of Extraction (Estimated)	Condition of Extraction (Actual)
Induced I	Test	(0.2±10%)g/ mL	6.04g	30.20mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
	Control	N.A.	N.A.	15.00mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
Induced II	Test	(0.2±10%)g/ mL	6.02g	30.10mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
	Control	N.A.	N.A.	15.00mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
Challenge	Test	(0.2±10%)g/ mL	6.00g	30.00mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
	Control	N.A.	N.A.	15.00mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours

10.1.2. Non-Polar Extract

Phase	Group	Extraction Ratio	Weight of Test Item	Extraction Volume	Condition of Extraction (Estimated)	Condition of Extraction (Actual)
Induced I	Test	(0.2±10%)g/ mL	5.96g	29.80mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
	Control	N.A.	N.A.	15.00mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
Induced II	Test	(0.2±10%)g/ mL	6.05g	30.25mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
	Control	N.A.	N.A.	15.00mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
Challenge	Test	(0.2±10%)g/ mL	5.93g	29.65mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours
	Control	N.A.	N.A.	15.00mL	50±2℃; 72±2 hours	50.0~50.0℃; 72hours

10.2.Extraction Condition

10.2.1. Polar Extract

Group	Observation Point	Phase	Extraction State		
			Color	Clarity	Precipitate
Test	Before Extract	Induced I	colorless	clear	No
		Induced II	colorless	clear	No
		Challenge	colorless	clear	No
	After Extract	Induced I	colorless	clear	No
		Induced II	colorless	clear	No
		Challenge	colorless	clear	No
	Before Used	Induced I	colorless	clear	No
		Induced II	colorless	clear	No
		Challenge	colorless	clear	No
Control	Before Extract	Induced I	colorless	clear	No
		Induced II	colorless	clear	No
		Challenge	colorless	clear	No
	After Extract	Induced I	colorless	clear	No
		Induced II	colorless	clear	No
		Challenge	colorless	clear	No
	Before Used	Induced I	colorless	clear	No
		Induced II	colorless	clear	No
		Challenge	colorless	clear	No

10.2.2. Non-Polar Extract

Group	Observation Point	Phase	Extraction State		
			Color	Clarity	Precipitate
Test	Before Extract	Induced I	light yellow	clear	No
		Induced II	light yellow	clear	No
		Challenge	light yellow	clear	No
	After Extract	Induced I	light yellow	turbid	No
		Induced II	light yellow	turbid	No
		Challenge	light yellow	turbid	No
	Before Used	Induced I	light yellow	turbid	No
		Induced II	light yellow	turbid	No
		Challenge	light yellow	turbid	No
Control	Before Extract	Induced I	light yellow	clear	No
		Induced II	light yellow	clear	No
		Challenge	light yellow	clear	No
	After Extract	Induced I	light yellow	turbid	No
		Induced II	light yellow	turbid	No
		Challenge	light yellow	turbid	No
	Before Used	Induced I	light yellow	turbid	No
		Induced II	light yellow	turbid	No
		Challenge	light yellow	turbid	No

10.3. Animal Body Weight

10.3.1. Polar Extract

Total body weight				
Animal number	Group	Sex	Before (g)	After (g)
1	Control	Female	363.52	505.19
2	Control	Female	325.63	450.99
3	Control	Female	395.03	552.86
4	Control	Female	357.62	478.48
5	Control	Female	319.37	389.83
6	Test	Female	311.24	468.84
7	Test	Female	309.26	425.30
8	Test	Female	358.83	454.70
9	Test	Female	318.32	501.80
10	Test	Female	337.82	458.42
11	Test	Female	321.08	430.35
12	Test	Female	323.36	437.04
13	Test	Female	307.28	455.03
14	Test	Female	320.94	413.16
15	Test	Female	306.23	459.80

10.3.2. Non-Polar Extract

Total body weight				
Animal number	Group	Sex	Before (g)	After (g)
1	Control	Female	349.47	464.08
2	Control	Female	353.68	514.25
3	Control	Female	349.75	474.54
4	Control	Female	307.79	398.71
5	Control	Female	376.95	520.65
6	Test	Female	336.36	406.29
7	Test	Female	311.44	440.13
8	Test	Female	343.09	426.46
9	Test	Female	362.40	472.17
10	Test	Female	323.36	418.25
11	Test	Female	306.36	435.09
12	Test	Female	369.20	449.79
13	Test	Female	310.01	428.74
14	Test	Female	338.20	412.89
15	Test	Female	318.32	410.42

10.4. Values of Skin Reactions

10.4.1. Individual Values of Skin Reactions (Polar-Extract)

Animal number	Group	Magnusson and Kligman scale					
		24±2 hours			48±2 hours		
		Control side	Test side	Positive (+/-)	Control side	Test side	Positive (+/-)
1	Control	0	0	-	0	0	-
2	Control	0	0	-	0	0	-
3	Control	0	0	-	0	0	-
4	Control	0	0	-	0	0	-
5	Control	0	0	-	0	0	-
Control group mean score/ Frequency of positive (%)		0	0	0%	0	0	0%
6	Test	0	0	-	0	0	-
7	Test	0	0	-	0	0	-
8	Test	0	0	-	0	0	-
9	Test	0	0	-	0	0	-
10	Test	0	0	-	0	0	-
11	Test	0	0	-	0	0	-
12	Test	0	0	-	0	0	-
13	Test	0	0	-	0	0	-
14	Test	0	0	-	0	0	-
15	Test	0	0	-	0	0	-
Test group mean score/ Frequency of positive (%)		0	0	0%	0	0	0%

10.4.2. Individual Values of Skin Reactions (Non-Polar-Extract)

Animal number	Group	Magnusson and Kligman scale					
		24±2 hours			48±2 hours		
		Control side	Test side	Positive (+/-)	Control side	Test side	Positive (+/-)
1	Control	0	0	-	0	0	-
2	Control	0	0	-	0	0	-
3	Control	0	0	-	0	0	-
4	Control	0	0	-	0	0	-
5	Control	0	0	-	0	0	-
Control group mean score/ Frequency of positive (%)		0	0	0%	0	0	0%
6	Test	0	0	-	0	0	-
7	Test	0	0	-	0	0	-
8	Test	0	0	-	0	0	-
9	Test	0	0	-	0	0	-
10	Test	0	0	-	0	0	-
11	Test	0	0	-	0	0	-
12	Test	0	0	-	0	0	-
13	Test	0	0	-	0	0	-
14	Test	0	0	-	0	0	-
15	Test	0	0	-	0	0	-
Test group mean score/ Frequency of positive (%)		0	0	0%	0	0	0%

10.4.3. Clinical Observations

During the test (Jun. 30, 2023-Jul 24, 2023) Laboratory Animal Observation

Group	Control		Test	
	Polar-Extract	Non-Polar-Extract	Polar-Extract	Non-Polar-Extract
正常	5/5*	5/5*	10/10*	10/10*
異常	0/5 ⁺	0/5 ⁺	0/10 ⁺	0/10 ⁺
死亡	0/5 [#]	0/5 [#]	0/10 [#]	0/10 [#]

Remarks: During the test period, animal clinical observations were carried out every Monday to Friday, and the observation content included whether the appearance and behavior of the animals were normal.

* (number of animals with normal clinical observation/total number of animals in each group)

+ (number of animals with abnormal clinical observation/total number of animals in each group)

(Number of dead animals in clinical observation/total number of animals in each group)

10.4.4. The test animals were observed for mobility, mortality, other clinical signs in skin, fur, and occurrence of excretions. There were no clinical signs including mortality and morbidity related with the test item during the testing period.

10.5. Conclusion

Under the condition of this test, neither the control nor the test group showed significant skin response on the treated area. Thus, the results indicated that the polar and non-polar extract of 「RELANO WOUND SPRAY」 did not produce skin sensitization in guinea pigs.

11. Deviation Statement

There were no deviations from the approved study plan which were judged to have any impact on the validity of the data.

12.Study Documents and Archives

The study pertinent records to be maintained will include, but not limited to, the following objects:

- Signed Study Plan
- Material record (e.g. Test Item/ Control Item/ Specimen Information Form, Receiving Item Confirmation Form, and Controlled Item Request Form.)
- Raw data for experimental handling(e.g. IACUC approval certification, Laboratory Animal Quarantine Report, Laboratory Animal Receive Record, and Animal Room Environmental Monitoring data)
- Raw data of experimental results (e.g., Skin Sensitisation Test Procedure-II, Skin Sensitisation Test - Animal Body Weight Record, Individual Values of Skin Reactions-II, Laboratory Animal Observation Record, Cage Card, Photograph of the Observation, Autoclave Record-Water, Diet, Bedding and etc.)
- Correspondence between the CBSCMHAC and the Sponsor Application Form
- A copy of Final Report

The above-mentioned data and documents will be maintained in the archives at Animal Center of Chang Bing Show Chwan Memorial Hospital for at least 6 years after report finalization. The retention of test item will be maintained in the storage cabinet (SC-01) in the control area of the test item management room at room temperature. The storage period of the test item will be reserved for at least 6 years after the final report becomes effective or be withdrawn after the validity period of the test item.

13. References

- 13.1 Taiwan Food and Drug Administration -Good Laboratory Practice for Nonclinical Laboratory Studies. (2019)
- 13.2 The Organisation for Economic Co-operation and Development (OECD)- Principles of Good Laboratory Practice(GLP). (1997)
- 13.3 Council of Agriculture, Executive Yuan, Animal Care and Use Program (2018)
- 13.4 World Health Organization. (2002). WHO Manual on Animal Influenza Diagnosis and Surveillance.
- 13.5 The Organisation for Economic Co-operation and Development (OECD) Test Guidelines 406: Skin Sensitisation Guinea Pig Maximisation Test and Buehler Test. (2021)
- 13.6 International Organization for Standardization (ISO) 10993-12, Biological evaluation of medical devices - Part 12: Sample preparation and reference materials (2021).
- 13.7 International Organization for Standardization (ISO) 10993-10, Biological evaluation of medical devices - Part 10: Test for Skin Sensitisation (2021).
- 13.8 Magnusson B. and Kligman A.M. (1970). Allergic Contact Dermatitis in the Guinea Pig. Charles G. Thomas; Springfield, Illinois.
- 13.9 SOP GLP-AF-00-09 Laboratory Animal Purchasing, Receiving, Quarantine, Domestication procedures and management
- 13.10 SOP GLP-AF-00-05 Animal Room Environmental Monitoring
- 13.11 SOP GLP-EQ-20-01 Microbalance (PA224C)
- 13.12 SOP GLP-EQ-20-06 Balance (PGW3502E)
- 13.13 SOP GLP-EQ-10-01 Biological safety cabinet (AC2-4S9-NS)
- 13.14 SOP GLP-EQ-20-05 Incubator (LM-570RD)
- 13.15 SOP GLP-TM-10-02 Skin Sensitisation Test (Guinea Pig Maximisation Test(GPMT))
- 13.16 SOP GLP-TM-10-03 Test Item Extraction

Appendix 1

Reliability Check

The positive control study (Study no. TW006-23015L01) was finished on May 15, 2023. According to the ISO10993-10 (2021) and OECD 406 Skin Sensitization (2021) guidelines, positive control shall be performed at least once every six months. 2-Mercaptobenzothiazole (2MBT) (Sigma, M3302-5G, Lot No 0000167089) were used for positive control substances, which was a suggested positive control substances according to ISO10993-10 (2021). The method of induction phase I and phase II for positive control assay was identical to the method described above in the study, and the method of challenge phase followed ISO10993-10 (2021) section 6.5.4.3.3 option b),2) to determine if any skin reaction observed in the test animals is due to sensitization or potential irritation. Respectively, the induction phase I, induction phase II and challenge phase were corresponding to 1%, 25% and 15% in dimethyl sulfoxide absolute (w/v) was used.

The frequency of positive challenge result was 80% at 24±2 hours and 70% 48±2 hours in the test group; moreover the score were 1 to 3 at 24±2 hours and 48±2 hours. Animals in the positive control group exhibited discrete or patchy erythema to intense erythema and swelling at the test site, therefore the test result showed that the response of the test item was positive, and validated test system reproducibility and sensitivity in laboratory.

Animal number	Group	Magnusson and Kligman scale					
		24±2 hours			48±2 hours		
		Control side	Test side	Positive (+/-)	Control side	Test side	Positive (+/-)
1	Control	0	0	-	0	0	-
2	Control	0	1	+	0	0	-
3	Control	1	1	-	1	1	-
4	Control	0	0	-	0	0	-
5	Control	0	0	-	0	1	+
Control group mean score/ Frequency of positive (%)		0.2	0.4	20%	0.2	0.4	20%
6	Test	0	1	+	1	1	-
7	Test	0	1	+	0	1	+
8	Test	0	0	-	0	1	+
9	Test	0	1	+	1	1	-
10	Test	0	2	+	0	2	+
11	Test	0	2	+	0	1	+
12	Test	1	3	+	1	3	+
13	Test	2	3	+	1	3	+
14	Test	0	1	+	1	2	+
15	Test	0	0	-	0	0	-
Test group mean score/ Frequency of positive (%)		0.3	1.4	80%	0.5	1.5	70%

Appendix 2

試驗物質/對照物質/檢體資料表(英文)

Test Item/ Control Item/ Specimen Information Form (English)

Information marked with "*" will be written in the final report, please fill in carefully.

		Test/Control ItemNo. TW025-23021L01-LA7512305003(1-12)	
It will be labeled by CBSCMHAC receiving personnel.			
Sponsor information			
*Sponsor/Company Name	Nanoplus Life Biomedical Technology Co., Ltd.		
*Sponsor Address	9 F., No. 179, Sec. 2, Tiding Blvd., Neihu Dist., Taipei City 114735, Taiwan (R.O.C.)		
*Sponsor's Representative	Tzu-Ying Hung (洪慈營)		
*Contact Person	Tzu-Ying Hung (洪慈營)	Telephone Number	(02)2656-0555
		E-mail	Cecilia@nanoplustech.com
*Test Name	RELANO WOUND SPRAY (良諾液態保護噴劑)		
*Estimated Transfer Date	06/15/2023 (mm/dd/yyyy)		
Transfer Method	☒ Personal delivery ☐ Post ☐ By sea/shipping ☐ Others :		
Transfer Conditions	☒ Room Temperature ☐ Refrigerated ☐ Freezing ☐ Dry ice ☐ Liquid Nitrogen ☐ Others :		

■ TEST ITEM/ □ CONTROL ITEM

Information of Test Item/ Control Item	
* Name of Test Item/Control Item	RELANO WOUND SPRAY (良諾液態保護噴劑)
* Batch/Lot Number	LOT51223050003
Type	☒ Medical Devices - Category: Class I ☒ Surface Device ☐ External Communicating Device: Circulating Blood : ☐ Yes/ ☐ No ☐ Implant Device ☐ Gas Pathways ☐ Other Type : ☐ Food; ☐ Cosmetics Products; ☐ Industrial Chemicals; ☐ Pesticide Products; ☐ Drug; ☐ Others :
Packing Condition	☐ In Bulk; ☐ Intact Packing; (☒ Bottled ☐ Boxed ☐ Tube ☐ Barrel ☐ Bag ☐ Others :)
Amount/Size	Test: 6 bot * 50±2 ml (1 g/1 mL); Retention: 6 bot * 50±2 ml (1 g/1 mL)
Manufacture Date	05/26/2023 (mm/dd/yyyy) ☐ No information provided.
Expiry Date/Use by	05/26/2025 (mm/dd/yyyy)
* Major Components	Nano Ion Water, Carboxymethylcellulose
* Purity/Concentration	Purity: NA Concentration: NA

試驗物質/對照物質/檢體資料表(英文)

Test Item/ Control Item/ Specimen Information Form (English)

*Description (Name)	External features: ☐ Regular; ☐ Irregular; ☒ Liquid; ☐ Powder; ☐ Granule; ☐ Flat; ☐ Other: _____
	Color: <u>Colorless</u>
	Stability: <u>N.A.</u>
	PH values: <u>N.A.</u>
	Sterilization: Has been Sterilized ☒ NO ☐ YES. If Yes, Please Select the Following Method ☐ EO Sterilization ☐ Gamma Sterilization ☐ Steam Sterilization ☐ Other _____ Thickness: _____ mm/pcs or ☒ Not Available. Surface Area: ☐ _____ cm ² /pcs ☐ Double Side ☐ Single Side ☐ All Side or ☒ Not Available. *According to FDA guidance, determine the appropriate amount of test material by using surface area to extractant volume ratios. Mass to extractant volume ratios should only be used if surface area cannot be calculated, or if use of mass will result in a larger sample. In addition, the test will be performed directly with the surface area provided by the sponsor.
* Storage Condition	☒ Room Temperature ☐ 2-8℃ ☐ -20±10℃ ☐ -65~-85℃ ☐ Liquid Nitrogen ☐ Keep in Dark Place ☐ Avoid Moisture ☐ Others: _____
Cutor/Not (Notes)	☐ Yes ☐ No ☒ N.A. (Liquid, Gel, Powder)
*Solvent (Solubility)	Solvent: <u>N.A.</u> Solubility: <u>N.A.</u>
Test Conditions (Dilution)	☒ No Dilution ☐ Yes, Test final concentration (%): _____; Recommended Solvent: _____
Item Return (Name)	☒ No, The remaining materials considered as retention? ☐ Yes ☒ No ☐ N.A. (Processing Method: _____)
	☐ Yes, only for un-used test item + Delivery Conditions: ☐ Room Temperature ☐ Refrigerated ☐ Frozen ☐ Dry Ice ☐ Liquid Nitrogen Recipient: _____, Telephone: _____ Address: _____
Attachment (Name)	☐ Certificate of Analysis ☐ Material Safety Data Sheet ☐ Stability Test Result ☐ Product Manual ☐ Others: _____ ☒ No Attachment
Extraction	Ratio: ☐ No Extraction ☐ By Surface Area: ☐ (3±10%)cm ² /mL ☐ (6±10%)cm ² /mL ☒ By Weight: ☐ (0.1±10%)g/mL ☒ (0.2±10%)g/mL ☐ Others: <u>According to the laboratory recommended sampling volume (weight) testing</u> Condition: ☐ (37±1)℃ (24±2)Hours ☒ (50±2)℃ (72±2)Hours ☐ Other: _____
Others	For Skin Sensitisation Test: Guinea Pig Maximisation Test

試驗物質/對照物質/檢體資料表(英文)

Test Item/ Control Item/ Specimen Information Form (English)

Precautions

- ◆ Note1. All information in the form including the blanks which can't be provided are disclosure and should taken responsibility by the sponsor.
 - ◆ Note2. If the test follow GLP, should provide extra amount of the test/control item with the same lot for retention. If sponsor doesn't provide the retention of test item/control item, the retention of a reserved test item/control item from each lot/batch of test item /control item is the responsibility of the Sponsor.(Example: 1mg/1 mL*1 bot or 1x10⁶ cell/1 mL*2 vial)
 - ◆ Note3. For retention, if the effective period is less than CBSCMHAC SOP definition, the test item/control item will be retained till the expiry date. If the expiry date is longer than CBSCMHAC SOP definition, the test item/control item will be retained for the period of CBSCMHAC SOP definition only. If the expiry date remained incomplete, it mentions that sponsor will agree that test facility will determine the earliest date, e.g. Exp Date: 2021(YYYY), sponsor didn't identify the MM/DD, the expiration date should be 01/01/2021 (means Jan. 01, 2021)
 - ◆ Note4. Sponsor should determinate, document and confirm the identity, strength, purity, stability, composition, method of synthesis, fabrication, derivation or other characteristics of the test item/control item before study. If the sponsor cannot provide the information, determination and documentation of the test item/control item are the responsibility of the Sponsor.
 - ◆ Note5. The test item/control item which has been destroyed or cutting will be discarded after the end of experiment unless applicant/sponsor have extra requirement. If the sponsor does not agree or have special requirements, should take the initiative to inform test facility and state the application prior to test. If related losses incurred because applicant/sponsor didn't inform, we will not be liable.
 - ◆ Note6. If sponsor wants to return the sample, the retention, inconsistencies and biological safety related issue of test item/control item are the responsibility of the Sponsor.
 - ◆ Note7. Note "N/A" or "N.A" if not applicable. Do not leave blank.
 - ◆ Note8. Test item and control item should be filled individually in "INFORMATION OF TEST ITEM/CONTROL ITEM".
 - ◆ Note9. GLP test : (1) Corrections and additions to a final report should be in the form of amendments. Amendments should clearly specify the reason for the corrections or additions. Lot number, test item photos and raw data cannot be amended by sponsor's requirement. (2) One protocol only can generate one report except for translation version. If the report with more than two languages version, we only issue English version protocol. We only issue amendment or additional language GLP report within three years and non-GLP report within one year. (3) The GLP compliance statement will state that we follow TFDA GLP and TAF OECD GLP norm unless sponsor's requirement. Chang Bing Show Chwan Memorial Hospital Experimental Animal Center (CBSCMHAC) have acquired Good Laboratory Practice Statement of Compliance from TFDA and TAF.
 - ◆ Note10. Please write down if carefully and in detail. "INFORMATION OF TEST ITEM/CONTROL ITEM" will be placed in the protocol and report along with a copy of this official document. If the information is not clear, we will exclude them from GLP statement.
 - ◆ Note11. The results shown in this test report refer only to the test item(s) tested.
 - ◆ Note12. If the sponsor is different from the applicant mentioned in the application form, the sponsor agrees that the study-related information, test/control item and application form provided or reply to CBSCMHAC shall be agreed by the sponsor. The report submitted by SGS is only to the applicant.
 - ◆ Note13. For commissioned tests under GLP specifications, each batch of test item/control item shall comply with the regulations of the planned regulatory agency (for example: FDA/OECD GLP). The test consignee should attach the relevant certificate of the test item. If it cannot be provided, it will be excluded from the GLP statement.
- CBSCMHAC makes no GLP compliance claim for the following items:
- ◆ 1. Sections 6.2.2, 6.2.3, 6.2.4, 6.2.5 in Taiwan Food and Drug Administration -Good Laboratory Practice for Nonclinical Laboratory Studies and The Organization for Economic Co-operation and Development (OECD)- Principles of Good Laboratory Practice(GLP). Characterization and verification of the test article identity and properties provided from sponsor, which are the responsibility of the sponsor.
 - ◆ 2. Sections 6.2.6 in Taiwan Food and Drug Administration -Good Laboratory Practice for Nonclinical Laboratory Studies and The Organization for Economic Co-operation and Development (OECD)- Principles of Good Laboratory Practice (GLP). A sample for analytical purposes from each batch of test item should be retained for all studies except short-term studies. Sponsor is responsible for the retention of samples for storage conditions other than room temperature and 4°C refrigeration.

Sponsor's Representative Signature/Date:

陳亭秀 08/18/2023

Appendix 3







秀傳醫療財團法人彰濱秀傳紀念醫院
Chang Bing Show Chwan Memorial Hospital

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